

```

37 • SELECT d.name AS Department,
38       p.name AS Head_Physician
39 FROM department d
40 JOIN physician p
41 ON d.head = p.employeeid;
42
43 SELECT *
44 FROM physician
45 WHERE position = 'Surgical Attending Physician';
46
47 SELECT *
48 FROM physician
49 WHERE name LIKE 'John%';
50
51 • SELECT COUNT(*) AS Total_Physicians
52 FROM physician;
53

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

employeeid	name	position	ssn
3	Christopher Turk	Surgical Attending Physician	333333333
6	Todd Quinlan	Surgical Attending Physician	666666666
7	John Wen	Surgical Attending Physician	777777777
NULL	NULL	NULL	NULL

physician 3 ×

Output

Action Output

#	Time	Action	Message
3	15:12:00	DROP TABLE IF EXISTS department	0 row(s) affected
4	15:12:03	DROP TABLE IF EXISTS physician	0 row(s) affected
5	15:12:06	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR...	0 row(s) affected
6	15:12:09	INSERT INTO physician VALUES (1, 'John Dorian', 'Staff Internist', '111111111'), (2, 'Elliot Reid', 'Attending Ph...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
7	15:12:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
8	15:12:18	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, 'Psychiatry', 9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9	15:12:24	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned

```

61
62 • SELECT position,
63       COUNT(*) AS Total_Employees
64 FROM physician
65 GROUP BY position
66 HAVING COUNT(*) > 1;
67
68 SELECT *
69 FROM physician
70 ORDER BY name ASC;
71

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

position	Total_Employees
Surgical Attending Physician	3

Result 5 ×

Output

Action Output

#	Time	Action	Message
5	15:12:06	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), position VARCHAR...	0 row(s) affected
6	15:12:09	INSERT INTO physician VALUES (1, 'John Dorian', 'Staff Internist', '111111111'), (2, 'Elliot Reid', 'Attending Ph...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
7	15:12:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
8	15:12:18	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, 'Psychiatry', 9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9	15:12:24	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
10	15:12:30	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
11	15:15:45	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	2 row(s) returned
12	15:15:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned

```

122
123 • SELECT *
124 FROM department d
125 WHERE EXISTS (

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employeeid	name	position	ssn
▶	5	Bob Kelso	Head Chief of Medicine	555555555
	3	Christopher Turk	Surgical Attending Physician	333333333
	2	Elliot Reid	Attending Physician	222222222
	1	John Dorian	Staff Internist	111111111
	7	John Wen	Surgical Attending Physician	777777777

physician 6 x

Output

Action Output

#	Time	Action	Message
✓	6 15:12:09	INSERT INTO physician VALUES (1, 'John Dorian', 'Staff Internist', '111111111'), (2, 'Elliot Reid', 'Attending Ph...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
✓	7 15:12:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓	8 15:12:18	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, 'Psychiatry', 9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	9 15:12:24	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓	10 15:12:30	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓	11 15:15:45	SELECT * FROM physician WHERE name LIKE 'John%'; LIMIT 0, 1000	2 row(s) returned
✓	12 15:15:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓	13 15:16:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned

```

123 • SELECT *
124 FROM department d
125 WHERE EXISTS (
126     SELECT 1
127     FROM physician p
128     WHERE p.employeeid = d.head

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	departmentid	name	head
▶	1	General Medicine	4
	2	Surgery	7
	3	Psychiatry	9
•	NULL	NULL	NULL

Result Grid

department 7 x

Apply

Output

Action Output

#	Time	Action	Message
✓	7 15:12:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
✓	8 15:12:18	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, 'Psychiatry', 9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✓	9 15:12:24	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON ...	3 row(s) returned
✓	10 15:12:30	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
✓	11 15:15:45	SELECT * FROM physician WHERE name LIKE 'John%'; LIMIT 0, 1000	2 row(s) returned
✓	12 15:15:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT...	1 row(s) returned
✓	13 15:16:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
✓	14 15:17:06	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid...	3 row(s) returned

```

123 • SELECT *
124 FROM department d
125 WHERE EXISTS (
126     SELECT 1
127     FROM physician p
128     WHERE p.employeeid = d.head

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Contents: |

departmentid	name	head
1	General Medicine	4
2	Surgery	7
3	Psychiatry	9
4		
5		
6		
7		
8		
9		
10		

department 7 x

Output

Action Output

#	Time	Action	Message
7	15:12:14	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT)	0 row(s) affected
8	15:12:18	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, 'Psychiatry', 9)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
9	15:12:24	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d.head = p.employeeid	3 row(s) returned
10	15:12:30	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
11	15:15:45	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	2 row(s) returned
12	15:15:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT(*) > 1	1 row(s) returned
13	15:16:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
14	15:17:06	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid = d.head)	3 row(s) returned

```

131 • SELECT MAX(employeeid) AS Maximum_Employee_ID
132 FROM physician;
133
134 • SELECT MIN(employeeid) AS Minimum_Employee_ID

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

Minimum_Employee_ID
1

Result 10 x

Output

Action Output

#	Time	Action	Message
10	15:12:30	SELECT * FROM physician WHERE position = 'Surgical Attending Physician' LIMIT 0, 1000	3 row(s) returned
11	15:15:45	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	2 row(s) returned
12	15:15:52	SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT(*) > 1	1 row(s) returned
13	15:16:37	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned
14	15:17:06	SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid = d.head)	3 row(s) returned
15	15:17:19	SELECT MAX(employeeid) AS Maximum_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned
16	15:17:42	SELECT MAX(employeeid) AS Maximum_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned
17	15:19:00	SELECT MIN(employeeid) AS Minimum_Employee_ID FROM physician LIMIT 0, 1000	1 row(s) returned